Observed Booking Patterns and Vehicle Popularity in an Indian Car Rental Platform: An Exploratory Analysis with Baseline Models

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Abstract

Car rental operators must decide how many vehicles, and which models, to place in each city, yet smaller operators often lack the data volume that sophisticated forecasting systems assume. This paper presents an exploratory, baseline-first analysis of observed bookings: 1,606 records from an Indian car rental company covering 31 cities between April and October 2023. After cleaning and normalising vehicle names, we characterise the geographic concentration of bookings, the distribution of booking prices and the popularity of vehicle models. We then fit a deliberately simple linear baseline of price on booking timestamp and illustrate a popularity-based vehicle recommender. Bookings are highly concentrated: Bengaluru accounts for roughly 39% of the total and the three largest cities for about two-thirds, while a small set of compact hatchbacks dominates vehicle popularity. The time-only price model yields a near-flat fit (slope ≈ −Rs 3 per day) and a test RMSE of ≈ Rs 4,057, comparable to the mean price of Rs 3,499, which suggests limited predictive value; confirming this against a mean-prediction benchmark is left for future work. We discuss why observed bookings are an imperfect proxy for demand and outline a validation-focused path towards forecasting models.

Keywords: car rental; booking data; vehicle popularity; exploratory data analysis; baseline models; India

1. Introduction

Car rental operators face a recurring planning problem: how many vehicles, of which types, should be available in each city, and when. The problem spans strategic decisions about fleet size and mix and tactical decisions about the daily deployment of vehicles across locations (Pachon et al., 2003; Oliveira et al., 2017). It has long been tied to revenue management. National Car Rental's return to profitability after adopting a revenue management program built on analytic models for capacity, pricing and reservations is a well-known example (Geraghty & Johnson, 1997), and later work has refined booking control for rentals of uncertain length (Li & Pang, 2017). All of these approaches depend on a clear picture of where, when and for which vehicles customers actually book.

Much of the published work we reviewed is framed around large operators in North America and Europe with extensive transaction histories. Smaller operators, and those in emerging markets, often start from far more modest booking logs. For them, a practical first question is not which sophisticated model to deploy, but what structure the data contain and which simple baselines any later model must beat (Hyndman & Athanasopoulos, 2021).

This paper takes that baseline-first view. Using 1,606 observed bookings from an Indian car rental company across 31 cities between April and October 2023, we make three contributions:

1. We characterise the geographic, price and vehicle-level structure of the observed bookings, and document a cleaning procedure for noisy vehicle names.
2. We fit a transparent time-only price baseline and report its test error, and we illustrate a popularity-based vehicle recommender as a reference rule for later work. The recommender is illustrative and has not yet been evaluated on held-out bookings.

3. We state explicitly what observed bookings can and cannot tell us about demand, and outline a validation-focused path to stronger forecasting models.

Section 2 reviews related work, Section 3 describes the data and methods, Section 4 presents results, Sections 5 and 6 discuss implications and limitations, and Section 7 concludes.

2. Related Work

Fleet and revenue management. Pachon et al. (2003) synthesise tactical fleet planning models, that is, the deployment of vehicles across rental locations. Oliveira et al. (2017) review the car rental fleet problem and integrate operational fleet issues with revenue management in a conceptual framework. Car rental differs from other revenue-management settings because rentals have variable lengths and vehicles can be moved between locations (Li & Pang, 2017; Oliveira et al., 2017). In such models demand forecasts are an input, which makes the quality of the underlying booking data central. Early industry experience with product management in car rental is described by Carroll and Grimes (1995), and Talluri and van Ryzin (2004) provide the general revenue-management background.

Forecasting and baselines. Tree ensembles such as random forests (Breiman, 2001) and gradient boosting (Chen & Guestrin, 2016) are popular for tabular prediction. Their benefit should be established against simple benchmarks and with time-respecting validation, in which models are tested on periods later than the training data (Bergmeir & Benítez, 2012; Hyndman & Athanasopoulos, 2021). Scale-dependent accuracy measures such as RMSE and MAE are easy to interpret but can mislead when series differ in scale, and scale-free alternatives are recommended for comparisons (Hyndman & Koehler, 2006).

Observed bookings as censored demand. Bookings record only the demand that the operator was able to serve. Revenue-management research addresses the gap between observed sales and underlying demand through “unconstraining” methods, which are particularly delicate when demand is small and sparse (Talluri & van Ryzin, 2004; Kourentzes et al., 2019).

Popularity baselines. Non-personalised rules that recommend what is most frequently chosen are a standard reference point in the recommender-systems literature (Ricci et al., 2015) and are appropriate when per-customer history is unavailable, as in our data.

Positioning. The models above generally take demand as given and focus on optimisation or control. Less visible in the work we reviewed, which is not exhaustive, is a descriptive account of what a modest booking log supports before such models are applied: where bookings concentrate, how prices are distributed, and which simple baselines are available and what they achieve. This paper provides such an account for one Indian operator. We do not claim that this setting is under-researched, only that a transparent descriptive baseline is a useful starting point for later modelling.

3. Data and Methods

3.1 Data

The data are an extract of booking records supplied by an Indian car rental company during an internship with the company and with its permission for academic use. The extract contains 1,606 bookings from 31 cities, with booking timestamps running from mid-April to mid-October 2023 (approximately 183 days). We retained the seven fields listed in Table 1 from the raw file. None of them identifies an individual customer.

Table 1. Fields retained from the booking extract.

Field	Description	Type
SecurityDeposit	Security deposit for the booking (Rs)	Numeric
City	City of the booking	Categorical
CarName	Vehicle model, as recorded	Categorical
StartTime	Time of day at which the rental starts	Text
Package Selected	Kilometre package chosen (e.g., “Unlimited KMs”, “60 KMs FREE”)	Categorical
price	Booking price (Rs)	Numeric
DateOfBooking	Booking timestamp (Unix epoch, milliseconds)	Numeric

Data preparation. The raw file was read with pandas, which skipped any rows it flagged as malformed (the number skipped was not recorded). Beyond this and the vehicle-name normalisation in Section 3.2, no rows were filtered, de-duplicated or imputed, and prices are used as recorded, including extreme values. We treat each row as one booking. A row is evidence of a booking made on the platform, not necessarily of a completed rental. The extract does not state whether price is a quoted or a final charged amount, or whether cancelled bookings are included; we list this as a limitation.

3.2 Preprocessing

Vehicle names were recorded inconsistently, often combining model, trim and year (e.g., “Brio 2017”, “I20 Asta CRDI”). We applied a rule-based normalisation: if a name contains a digit, we keep the first token (capitalised) and any middle tokens but drop the final token, which usually carries a year or trim designation; otherwise the name is left unchanged. The two examples above become “Brio” and “I20 Asta”. The rule trades some trim information for consistency and is heuristic. Residual variants remain, for example upper-case entries such as “BALENO SIGMA”, names that reduce to a brand only, and duplicates such as “MARUTI SWIFT” alongside “Swift” and “TATA NEXON” alongside “Nexon”. Counts for the affected models are therefore slightly understated, and the close ranking of Kwid and Swift in Section 4.3 should be read with caution. Figure 3 shows about 50 distinct labels after cleaning and pooling; a manually curated mapping to canonical model names is listed as future work. For the visualisation of vehicle popularity only, models with fewer than five bookings were pooled into an “Others” category.

3.3 Exploratory analysis

We tabulated bookings by city and by vehicle model and plotted price by city to identify concentration, dispersion and extreme values. No outliers were removed.

3.4 Baseline: price versus booking timestamp

To test whether time alone explains price, we fitted an ordinary least squares regression (scikit-learn; Pedregosa et al., 2011) of price on the booking timestamp. The model is deliberately time-only. The other retained fields (city, vehicle, kilometre package, security deposit and start time) are natural next predictors but were not used, because the aim was to test a pure time trend. The data were split at random into an 80% training set (1,284 bookings) and a 20% test set (322 bookings), using a fixed random seed of 42. We report the test-set mean squared error (MSE), whose square root (RMSE) is expressed in rupees, together with the fitted slope. We did not fit a training-mean benchmark, report MAE or R^2, or use a chronological split; these are the main gaps in the evaluation and are addressed in Sections 6 and 7.

3.5 Illustrative baseline: popularity-based recommender

For each city c, the recommender returns the vehicle model with the most bookings in that city: $\hat{y}(c) = \text{argmax}_m\, n(c, m)$, where $n(c, m)$ is the number of bookings of model m in city c. Cities absent from the data return a "no data" message. The rule needs no training and serves as the reference against which any personalised or time-aware recommender should be compared. Its rankings were computed on all bookings, so no held-out accuracy can be reported; we therefore present it as an illustration rather than an evaluated model.

4. Results

4.1 Geographic concentration

Bookings are heavily concentrated (Figure 1). Bengaluru accounts for roughly 630 of the 1,606 bookings (≈39%), followed by Mumbai (≈240) and Delhi (≈195); together these three cities contribute about two-thirds of all bookings. Pune, Chennai, Noida and Hyderabad form a middle tier with roughly 65–110 bookings each, while about two-thirds of the 31 cities have fewer than 20 bookings. Any city-level estimate for these thin cities will therefore be unreliable.

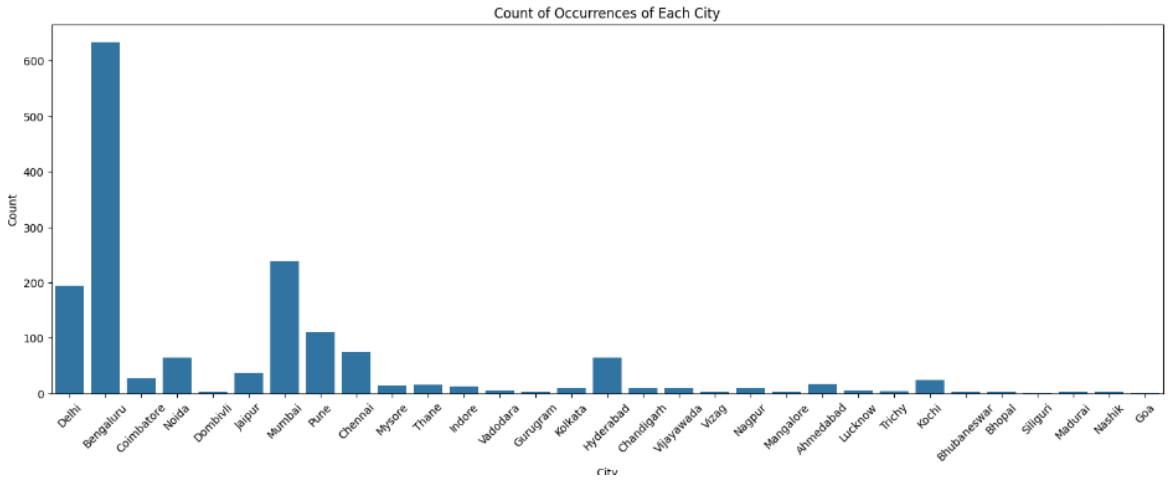


Figure 1. Number of bookings per city (31 cities, n = 1,606), in order of first appearance in the data.

4.2 Price distribution

Prices are right-skewed (Figure 2). The mean booking price is Rs 3,498.86 and most bookings fall below about Rs 10,000, but isolated bookings exceed Rs 25,000, the largest (about Rs 41,000) in Delhi. The widest price ranges appear in the cities with the most bookings (Delhi, Bengaluru, Mumbai and Pune), which partly reflects sample size: more observations make extreme values more likely. Extreme values were retained. Whether they reflect long rentals, premium vehicles or recording errors cannot be determined from the retained fields.

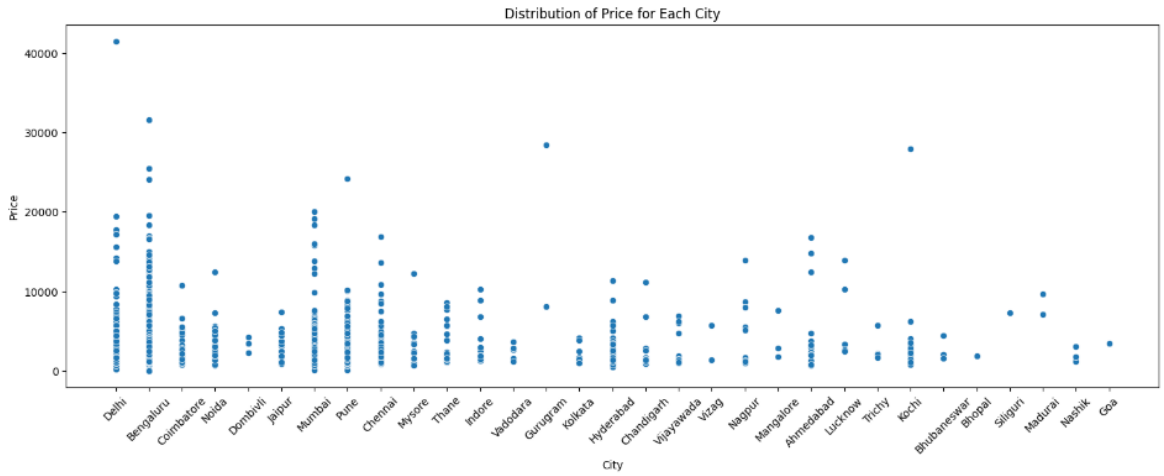


Figure 2. Distribution of booking price (Rs) for each city.

4.3 Vehicle popularity

After normalisation, bookings are concentrated in a small number of compact models (Figure 3). Kwid and Swift are the most frequently booked, with on the order of 160–170 bookings each and roughly one-fifth of all bookings between them, followed by models such as Alto, Baleno, Tiago and Eon (each with roughly 65–95 bookings). A long tail of rarely booked models, pooled as "Others" in the figure, accounts for about 8% of bookings. Section 3.2 describes the caveats that apply to these counts.

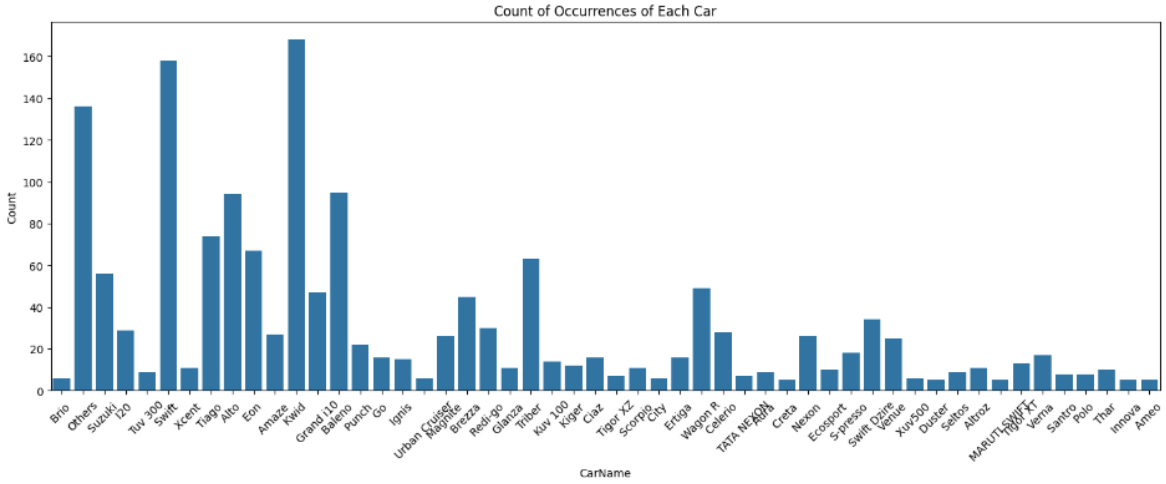


Figure 3. Number of bookings per vehicle model after normalisation; models with fewer than five bookings are pooled as "Others".

4.4 Price versus booking timestamp

The fitted slope is -3.47×10^{-8} Rs per millisecond, equivalent to about −Rs 3.0 per day, or roughly −Rs 550 across the 183-day window (Table 2, Figure 4). The test RMSE is about Rs 4,057, the same order as the full-sample mean price of Rs 3,499, reflecting the heavy right tail in Figure 2. The fitted line is nearly flat relative to the scatter of the points, which suggests that booking timestamp adds little to a constant typical price. We cannot confirm this, however: no training-mean benchmark was evaluated on the same test bookings, and the mean price quoted here is computed on the full sample, so the comparison is indicative only. We also did not test the significance of the slope and, given the dispersion and the extreme values, it should not be read as evidence of a price trend.

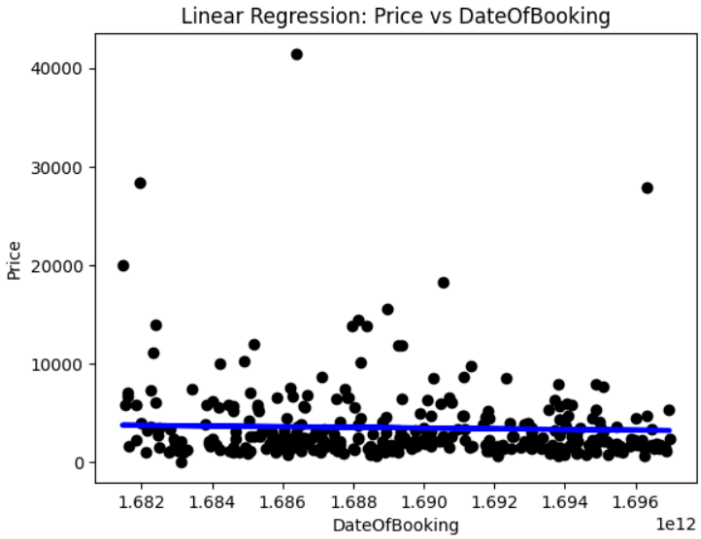


Figure 4. Linear regression of price on booking timestamp. The line was fitted on the training set and is drawn over the test observations. The horizontal axis shows Unix milliseconds: 1.682×10^{12} corresponds to 20 April 2023 and 1.696×10^{12} to 29 September 2023 (UTC).

Table 2. Summary of quantitative results.

Quantity	Value
Bookings analysed	1,606
Cities	31
Booking period	Mid-April to mid-October 2023 (≈183 days)
Train / test split	1,284 / 322 (random, seed 42)
Mean booking price (full sample)	Rs 3,498.86
Fitted slope, price on booking timestamp	-3.47×10^{-8} Rs per ms (≈ −Rs 3.0 per day)
Test MSE	16,456,858.7
Test RMSE	≈ Rs 4,056.7
Share of bookings, Bengaluru	≈ 39%
Share of bookings, three largest cities	≈ 66%

4.5 Popularity-based recommendation (illustration)

Applied to Delhi, the recommender returns Swift as the most frequently booked model in that city. Because the rule is computed per city, its output can differ from the platform-wide ranking in Figure 3 and adapts to local preferences. For the thin cities in Figure 1, however, the “most popular” model rests on only a handful of bookings and will be unstable.

5. Discussion

Operational reading. The concentration of bookings in a few cities and a few compact models suggests where planning effort pays off most: a small number of cities carry most of the volume, and a small group of vehicle models covers a large share of it. These are descriptive observations for one operator over six months, not causal claims, and we have not forecast bookings or sized a fleet. For the many thin cities, estimates should be pooled across cities or treated conservatively rather than computed city by city.

Why the time-only price model fits poorly. Rental prices plausibly depend on vehicle class, kilometre package, rental duration and city, none of which the baseline uses, and the retained fields do not include rental duration, so bookings of very different lengths are compared as if they were alike. We therefore hypothesise that the near-flat fit reflects an under-specified model rather than the absence of structure in prices. The result is still informative: if it is confirmed against a mean benchmark, it shows that a naive time trend is not a credible predictor, and that later models should include vehicle, package, city and duration.

Bookings are not demand. Observed bookings are limited by the vehicles the operator listed in each city and by availability at the time of search, so they understate unconstrained demand, particularly in cities with small fleets. In addition, the timestamp analysed is the booking timestamp, not the trip date. Our analysis therefore describes realised booking activity, and conclusions about seasonality or peak travel periods would require trip-date information.

6. Limitations

This is a single-operator study based on 1,606 observed bookings over roughly six months. The window is too short to identify annual seasonality or holiday effects, and the extract contains no weather, calendar or macroeconomic variables, so the influence of such factors is not examined. The evaluation has three main gaps. First, the price baseline is assessed with MSE and RMSE only, on a random rather than chronological split, and without a training-mean benchmark, MAE or R^2. Second, the recommender is illustrative: its city rankings were computed on all bookings, so no held-out accuracy, coverage or per-city performance is reported. Third, price is summarised by its mean and by figures only; its median and interquartile range are not reported. In addition, the extract does not state whether price is a quoted or final amount or whether cancellations are included, and vehicle-name normalisation is heuristic and leaves residual variants (Section 3.2).

We also prototyped a linear regression that predicted a label-encoded vehicle identifier from city code and booking timestamp. We do not report it as a result. Label encoding imposes an arbitrary numeric ordering on nominal categories, rounding a continuous prediction to a class index has no meaningful interpretation, and the prototype was fitted to the full sample without hold-out evaluation. The appropriate formulation is multiclass classification evaluated with top-k accuracy, as outlined below.

7. Conclusion and Future Work

Using 1,606 observed bookings from an Indian car rental company, we showed that booking activity is strongly concentrated in a few cities and a few compact vehicle models, that prices are heavy-tailed, and that booking timestamp alone appears to carry little information about price. A popularity-based recommender provides a transparent, training-free reference rule for vehicle popularity by city, although it has not yet been evaluated on held-out data. These results describe realised bookings rather than unconstrained demand, and they define the baselines that more elaborate forecasting or fleet-planning models should be required to beat on this kind of data.

Future work falls into six areas:

1. **Benchmarking and temporal validation.** Compare the regression with a training-mean predictor on the same chronological test period, reporting MAE, RMSE and R^2, and use rolling-origin evaluation as data allow (Bergmeir & Benítez, 2012).
2. **Evaluating the recommender.** Fit city rankings on training bookings only and report top-k accuracy, coverage and per-city results, with a global-popularity fallback for thin or unseen cities.
3. **Forecasting models.** Compare seasonal-naive benchmarks with random forests and gradient boosting using calendar and lagged features, reporting MAE, RMSE and a scale-free measure such as MASE (Hyndman & Koehler, 2006).

4. **Vehicle choice as classification.** Replace the prototype in Section 6 with multiclass classification, evaluated against the popularity baseline.
5. **Pooling across cities.** Use hierarchical or shrinkage estimators so that thin cities borrow strength from larger ones.
6. **Richer, cleaner data.** Curate a canonical vehicle-name mapping, document booking status and the meaning of price, and add trip dates, rental duration, weather, holidays and local events over a longer history, with estimates of unconstrained demand (Kourentzes et al., 2019).

Data Availability and Ethics

The booking data are proprietary to the company that supplied them and cannot be shared. They were used with the company's permission, and the retained fields contain no customer identifiers. Analysis was performed in Python using pandas, scikit-learn, matplotlib and seaborn, with a fixed random seed (42) for the train–test split.

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